

Problem Statement 01: Predictive Maintenance

Background

Naval and merchant ships depend on continuous propulsion and auxiliary machinery across harsh maritime environments. Unplanned mechanical breakdowns at sea compromise operational readiness, risk vessel safety, and inflict heavy maintenance and dockyard expenses.

Problem

Marine machinery operates under fluctuating engine loads (harbour idle, cruise, full power), masking early degradation patterns. Sensor streams frequently contain transient dropouts, noise, and missing entries. Maintenance crews need advance warning of both the time remaining before failure and the exact failing component.

Aim

Develop a machine learning model that ingests multi-sensor time-series logs (temperatures, pressures, vibrations, fuel rack metrics) to accurately predict the Remaining Useful Life (RUL in hours) and classify the specific failure mode (failure vs. no failure).

Output

1. Complete, reproducible training and inference source code (train.py, predict.py)
2. Exported model weights/checkpoints (.pt or .onnx)
3. Predictions CSV mapping unit_id to predicted failure probability and RUL

Data

File Structure:

PS1_public/

├── train.csv

├── test.csv

└─ sample_submission.csv

Data Description:

Column	Description	Type
unit_id	Machine identifier	int
timestamp	Sensor reading timestamp	datetime
temperature_c	Operating temperature (°C)	float
pressure_kpa	System pressure (kPa)	float
vibration_hz	Vibration frequency	float
motor_current_a	(Hz) Motor current	float
rotational_speed_rpm	(Amperes) Rotational speed (RPM)	int
torque_nm	Torque	float
target_failure	(Newton-meters) Will it fail? (0/1)	int
rul_hours	Hours until failure	float

Note: target_failure and rul_hours are only in train.csv

Submission Format

File:

teamname_PS1.csv

1. Unit_id, target_failure
2. 1,0.05
3. 2,0.95
4. 3,0.78

Requirements:

1. The submission file must contain exactly 2 columns: `unit_id` and `target_failure`.
2. `target_failure` must be a float value (0 or 1), representing the predicted class.
3. The file must contain the same number of rows as `test.csv`, with no missing values (NaN), and must be saved as `teamname_PS1.csv`.

Evaluation Metric

Your submission will be evaluated using the following classification metrics:

1. F1-Score: Measures the balance between precision and recall. Higher is better.
2. ROC-AUC: Measures the model's ability to distinguish between the positive and negative classes. Higher is better.

Final Score = $(0.7 \times \text{F1-Score}) + (0.3 \times \text{ROC-AUC})$